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Quantum reaction rate from higher derivatives of the thermal flux-flux autocorrelation function at time zero

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(Received 20 October 2004; accepted 2 November 2004; published online 7 January 2005)

A quantum theory of thermal reaction rates is presented which may be viewed as an extension of the recently developed “quantum instanton” (QI) model [W. H. Miller, Y. Zhao, M. Ceotto, and S. Yang, *J. Chem. Phys.* **119**, 1329 (2003)]. It is based on using higher derivatives of the flux-flux autocorrelation function $C(t)$ (as given by Miller, Schwartz, and Tromp) at $t=0$ to construct a short time approximation for $C(t)$. Tests of this theory on 1d and collinear reactions, both symmetric and asymmetric, show it to be more accurate than the original QI model, giving rate constants to $\sim 5\%$ for a wide range of temperature. © 2005 American Institute of Physics.
[DOI: 10.1063/1.1839177]

I. INTRODUCTION

The recently developed “quantum instanton” (QI) approximation¹ for calculating thermal rate constants of chemical reactions $k(T)$ shows considerable promise as a general method for obtaining reasonably accurate rates for reactions in complex molecular systems (i.e., those with many degrees of freedom), such as reactions in liquids, clusters, solids, or proteins. This is because the rate is expressed wholly in terms of the (quantum) Boltzmann operator $e^{-\beta\hat{H}}$ and properties thereof, which can be evaluated for complex systems using Monte Carlo path integral methods.² Test calculations for simple model problems indicate it to give rate constants accurate to at least $\pm 20\%$ over a wide range of temperature,^{1,3–7} from low temperature in the deep tunneling regime, to high temperature in the classical regime. Applications have been carried out for several collinear atom-diatom reactions, $A+BC\rightarrow AB+C$,^{1,3} and also such reactions in full 3d space,⁴ for the $H+CH_4\rightarrow H_2+CH_3$ in full 3d space,⁵ and for the Borgis *et al.* model of proton transfer in a polar solvent⁷ ($A+HB\rightarrow AH^++B^-$ in 225 methyl chloride molecules).

As always, though, one is looking for improvements in theoretical methodology, and such is the purpose of this paper. The approach presented herein, which can be viewed as an extension of the QI model, is more systematic (i.e., has fewer *ad hoc* ingredients), and tests on several model problems indicate it to be more accurate than the original QI model.

The primary “engine” for the present approach is the ability developed by Predescu and Doll⁸ (PD) for calculating *derivatives* of equilibrium path integral expressions. They initially applied their approach to calculate specific heats (essentially a second derivative of the partition function with respect to reciprocal temperature $\beta=1/k_B T$), but their methods can be readily applied to calculate zero time derivatives

of the flux-flux autocorrelation function of Miller, Schwartz, and Tromp,⁹

$$C(t) = \text{Tr}[e^{-\beta\hat{H}/2}\hat{F}e^{-\beta\hat{H}/2}e^{i\hat{H}t/\hbar}\hat{F}e^{-i\hat{H}t/\hbar}], \quad (1.1)$$

the integral of which gives the rate constant for a chemical reaction,

$$k(T) = Q_R(T)^{-1} \int_0^{+\infty} dt C(t), \quad (1.2)$$

where $Q_R(T)$ is the reactant partition function (per unit volume for a bimolecular reaction). Since $C(t)$ is an even function of time, the PD methodology can be used to obtain the even derivatives of $C(t)$ at $t=0$,

$$C^{[2n]}(0) = \left(\frac{d}{dt} \right)^{2n} C(t) \Big|_{t=0} \quad n = 1, 2, \dots \quad (1.3a)$$

The key feature of the PD approach is that the derivatives are evaluated by *finite difference*, but *inside* the path integral expression. This means that higher derivatives of the potential energy function are not required (which would be if one evaluated the derivatives analytically) and also that the zero time derivatives are obtained with the same order of statistical error as the zero time correlation function itself $C(0)$,

$$C(0) = \text{Tr}[e^{-\beta\hat{H}/2}\hat{F}e^{-\beta\hat{H}/2}\hat{F}]. \quad (1.3b)$$

We refer the reader to the original papers for more details of this methodology.⁸

The subject of this paper is how best to use this derivative information in order to obtain reaction rates. The zero time derivatives obviously allow a Taylor series expansion of $C(t)$ for short time,

$$C(t) = C(0)(1 + c_1 t^2 + c_2 t^4 + \dots), \quad (1.4a)$$

where

$$c_n = C^{[2n]}(0)/[(2n)!C(0)], \quad (1.4b)$$

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but this can clearly not be used directly in Eq. (1.2) to obtain the rate constant. The original QI model¹ can be viewed (at least in its simplest, one dividing surface version) as a second-order expansion,

$$C(t) = C(0)(1 + c_1 t^2), \quad (1.5a)$$

which is subjected to a cumulant resummation

$$1 + c_1 t^2 \rightarrow e^{c_1 t^2}, \quad (1.5b)$$

from which the rate constant is then given by Eq. (1.2) as (since c_1 is negative)

$$k(T) = Q_R^{-1} C(0) \frac{1}{2} \sqrt{\frac{\pi}{-c_1}}, \quad (1.5c)$$

or in the original QI notation

$$k(T) = Q_R^{-1} C(0) \frac{\hbar \sqrt{\pi}}{2\Delta H}, \quad (1.5d)$$

where ΔH is an energy variance

$$\Delta H = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} \quad (1.6)$$

with

$$\langle \hat{H}^k \rangle = \text{tr}[e^{-\beta \hat{H}/2} \hat{F} e^{-\beta \hat{H}/2} \hat{F} \hat{H}^k] / C(0), \quad (1.7)$$

$k=1, 2, \dots$ [Equation (1.7) for ΔH is not precisely that given by the original QI model, though very closely related to it; the rates given by the original QI expression are somewhat more accurate than those given with the ΔH of Eq. (1.7).¹] The PD methodology, however, is able to produce the fourth, sixth, etc., derivatives, and the goal of this paper is to find a practical way for utilizing this additional information to extend the QI model and obtain more accurate reaction rates.

To conclude this Introduction, we note that there is other work very much in the same spirit as the present, namely using various ways to construct short time approximations to the Miller-Schwartz-Tromp (MST) flux correlation function. This includes work by Yamashita and Miller,¹⁰ Thirumalai and Berne,¹¹ and most recently Rabani, Krilov, and Berne¹² and Sim, Krilov, and Berne¹³ on numerical analytic continuation of the imaginary time correlation function, and that of Hansen and Andersen¹⁴ which also uses a model for the short time behavior of $C(t)$.

II. USING HIGHER DERIVATIVES OF THE FLUX-FLUX CORRELATION FUNCTION

The most obvious way to utilize the higher derivatives of $C(t)$ is to subject the Taylor's series in Eq. (1.4) to a higher-order cumulant resummation,

$$C(t) = C(0) \exp(b_1 t^2 + b_2 t^4 + \dots). \quad (2.1)$$

If one has the N coefficients $c_1, \dots, c_N$ from Eq. (1.4), then there are standard procedures for generating the N values $b_1, \dots, b_N$ in Eq. (2.1). The (in general numerical) integral of $C(t)$ in Eq. (2.1) then gives the rate constant via Eq. (1.2).

We tried this procedure first, and it works fairly well at very low temperature (the simplest $N=1$ version, i.e., the Gaussian approximation of Eq. (1.5), is actually reasonably

accurate in the deep tunneling regime), but it completely fails at higher temperature. The reason for this is apparent when one recalls that the free particle correlation function (which is essentially the high temperature limit) has the form

$$C(t) = C(0) \left/ \left[1 + \left(\frac{2t}{\hbar\beta} \right)^2 \right]^{3/2} \right. \quad (2.2a)$$

or

$$C(t) = C(0) \exp \left\{ -\frac{3}{2} \ln \left[1 + \left(\frac{2t}{\hbar\beta} \right)^2 \right] \right\}. \quad (2.2b)$$

Expanding the logarithm gives the cumulant form for $C(t)$,

$$C(t) = C(0) \exp \left[-\frac{3}{2} \left(x - \frac{x^2}{2} + \frac{x^3}{3} + \dots \right) \right] \quad (2.3)$$

with $x = (2t/\hbar\beta)^2$. The power series expansion of $\ln(1+x)$, however, converges only for $|x| < 1$, i.e., for

$$t < \frac{\hbar\beta}{2},$$

which is too short to obtain the integral of $C(t)$ reliably.

After trying a variety of approaches for dealing with this problem, we settled on perhaps the simplest that one can imagine, namely augmenting the power series in the exponent of Eq. (2.1) with a log term: i.e., the ansatz for $C(t)$ is a “log-augmented cumulant” expansion,

$$C(t) = C(0) \exp \left\{ b_0 \ln \left[1 + \left(\frac{2t}{\hbar\beta} \right)^2 \right] + b_1 t^2 + \dots + b_N t^{2N} \right\}, \quad (2.4)$$

and one thus needs $(N+1)$ coefficients (i.e., derivatives) $c_1, \dots, c_{N+1}$ to determine $b_0, b_1, \dots, b_N$. We also explored the possibility of allowing the coefficient of t^2 in the argument of the log term to be a variable—i.e., $\ln(1+\lambda t^2)$ —but found little improvement from keeping it fixed at the free particle value $\lambda = (2/\hbar\beta)^2$.

In the results presented in the following section, we refer to the various values of N in Eq. (2.4) as the “ $(N+1)$ parameter model.” i.e., the two-parameter model ($N=1$) gives the exponent in Eq. (2.4) as

$$b_0 \ln \left[1 + \left(\frac{2t}{\hbar\beta} \right)^2 \right] + b_1 t^2, \quad (2.5)$$

and thus requires the two parameters c_1 and c_2 , i.e., the second and fourth derivatives of $C(t)$ at $t=0$, in order to determine the values of b_0 and b_1 . One may view the two-parameter model as the first step beyond the earlier QI approximation, which requires only the 2nd derivative (and is thus a “one-parameter model”). Since the t^2 term in the exponent does a reasonably good job of describing the deep tunneling regime, and the log term does so for the high temperature regime, the two-parameter model is capable, at least qualitatively, of describing a wide range of temperature.

Finally, we note that the specifics of determining the coefficients $b_0, \dots, b_N$ from the values of $c_1, \dots, c_{N+1}$, are the same as the standard cumulant approach: one expands both Eqs. (1.4) and (2.4) in power series in t^2 and equates the coefficients of like powers. i.e., one sets

$$\ln(1 + c_1 t^2 + \dots + c_{N+1} t^{2N+2}) = b_0 \ln \left[1 + \left(\frac{2t}{\hbar\beta} \right)^2 \right] + b_1 t^2 + \dots + b_N t^{2N}, \quad (2.6)$$

and expands the log terms on both sides in power series in t^2 . For the two-parameter model ($N=1$), for example, this leads to the following two equations:

$$c_1 = b_1 + b_0 \left(\frac{2}{\hbar\beta} \right)^2, \quad (2.7a)$$

$$c_2 - c_1^2/2 = -\frac{b_0}{2} \left(\frac{2}{\hbar\beta} \right)^4, \quad (2.7b)$$

with the solution

$$b_0 = \left(\frac{\hbar\beta}{2} \right)^4 (c_1^2 - 2c_2), \quad (2.8a)$$

$$b_1 = c_1 - \left(\frac{\hbar\beta}{2} \right)^2 (c_1^2 - 2c_2). \quad (2.8b)$$

III. ONE-DIMENSIONAL EXAMPLES

A. Symmetric barrier

We first apply the methodology of Sec. II to a standard one-dimensional model of a chemical reaction, namely the 1d Eckart potential barrier with parameters chosen to resemble the H+H₂ minimum energy reaction path: the potential function is

$$V(x) = V_0 \operatorname{sech}^2(ax) \quad (3.1)$$

with the parameters $V_0=0.425$ eV, $a=1.36$ a.u., and the mass $m=1060$ a.u. All matrix elements of the Boltzmann operator were calculated by diagonalizing the Hamiltonian in a discrete variable representation basis.¹⁵

Figure 1 shows the correlation functions [normalized to $C(0)$, its value at $t=0$] given by Eq. (2.4), with the parameters $\{b_n\}$ given by Eq. (2.7) and its higher-order versions. The results are shown for the two- and four-parameter models, compared to the exact correlation function, for a high [$T=2000$ K, Fig. 1(a)] and a low [$T=200$ K, Fig. 1(b)] temperature. One sees the effectiveness of the log term in the exponent of Eq. (2.4), especially at the higher temperature, as one would expect, but the description at low T is also very good.

Examining the values of the parameters $\{b_n\}$ from the various calculations shows the typical behavior $b_0 \gg b_1 \gg b_2 \sim b_3 \dots \sim b_{N-1}$, confirming the semiquantitative aspect of the two-parameter model. Adding higher-order terms, though, does improve the accuracy (see next figure). In some cases the parameter b_n for the highest power of t^2 in the exponent in Eq. (2.4) is positive, and though small, this leads to a divergent correlation function as $t \rightarrow +\infty$. In such cases this term in the cumulant expansion is omitted.

Figure 2 shows the percentage error in the rate constant given by the two-, three-, and four-parameter versions of Eq. (2.4) [and Eq. (1.2)], as a function of temperature. All three versions are accurate to 5% for temperature $T > 250$ K; at $T=200$ K they are all good to 10%, and the four-parameter

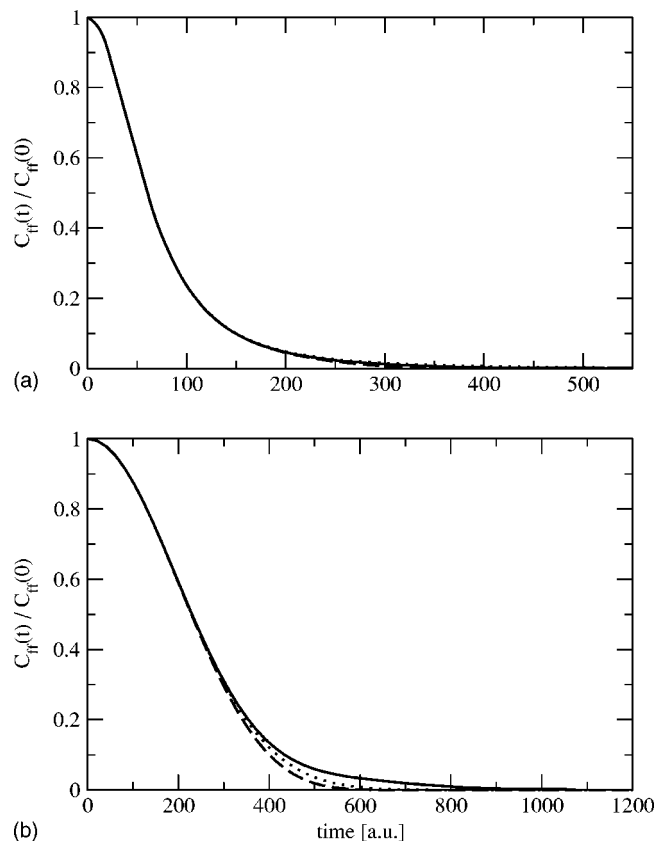


FIG. 1. Plot of the flux-flux autocorrelation function normalized with respect to its zero time value. The continuous line shows the exact correlation function, the dotted line the two-parameter model, and the dashed line the four-parameter model at (a) $T=2000$ K and (b) $T=200$ K.

version gives this level of accuracy also at the lowest temperature, $T=150$ K. The most encouraging feature of these results is how well the simplest (two-parameter) version of the approach works, and how stable the results are upon increasing the order of the log-augmented cumulant expansion.

B. Asymmetric barrier

We next consider the asymmetric Eckart barrier, for which the potential function is

$$V(x) = \frac{V_0(1-\alpha)}{1+e^{-2ax}} + \frac{V_0(1+\sqrt{\alpha})^2}{4 \cosh^2(ax)}, \quad (3.2)$$

where the parameters (a, m, V_0) were chosen to be the same as for the symmetric case and with α the asymmetry parameter. A value of $\alpha=1$ corresponds to the symmetric case; we have chosen $\alpha=1.25$, which gives a 25% difference in barrier height from the left (reactant) and right (product) side of the barrier.

This example raises the question of what should be the choice of the dividing surface (DS) for defining the flux operator in Eq. (1.1) *et seq*, since it is now not determined by symmetry. If the correlation function $C(t)$ were being calculated exactly, then its integral (i.e., the rate constant) would be invariant to the location of the DS.⁹ The short time approximation we are using in Sec. II [i.e., Eq. (2.4)], however,

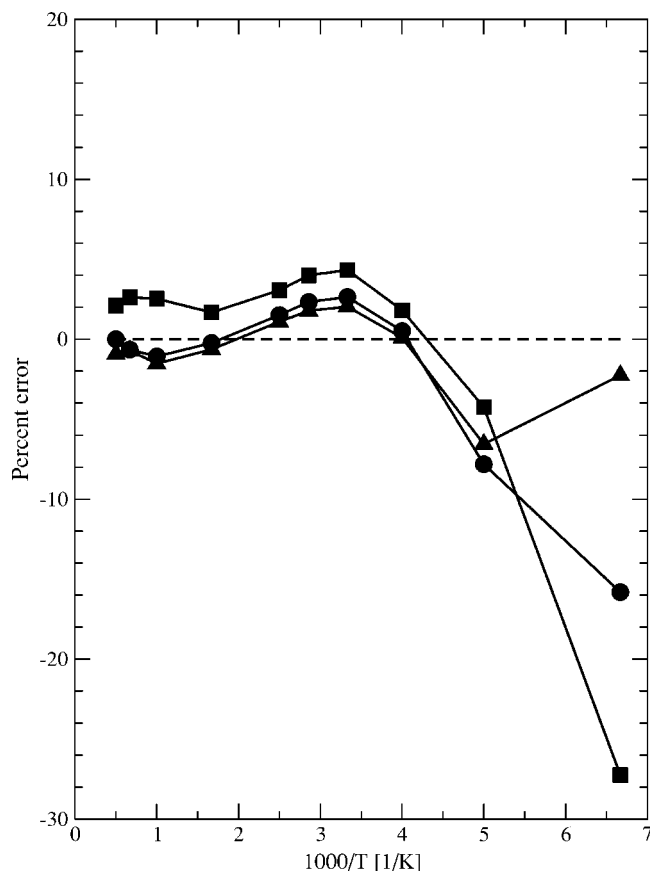


FIG. 2. Percentage error of the thermal rate constant vs temperature for different approximations in the case of the symmetric Eckhart potential: filled squares are the two-parameter model, circles the three-parameter model, and triangles the four-parameter model.

is obviously not exact, so the value of the rate constant given by its integral will depend on the choice of the DS. Since the log-augmented cumulant we use for $C(t)$ cannot describe recrossing effects in the correlation function (because this form is always positive)—i.e., it is a kind of transition state theory—one should choose the DS to minimize any such behavior. We have investigated a variety of criteria for choosing the optimum DS for this model, and the one that has performed best is the one which minimizes the coefficient of the highest-order term in the log-augmented cumulant expansion. In the two- (or three-) parameter version of the model, therefore, the DS is chosen to minimize the coefficient b_1 (or b_2). This criterion has the intuitive justification of making the cumulant expansion most rapidly convergent, but at present we can give no more fundamental support for this choice. (For the case of a symmetric reaction, this criterion produces the DS expected from symmetry.)

Figure 3 shows the value of this coefficient (that of the highest-order term in the cumulant expansion) as a function of the location of the dividing surface for the asymmetric Eckart barrier. The solid and dashed lines are for the two- and four-parameter models, respectively, and the upper panel is for a high temperature (1500 K) and the middle panel for a low temperature (150 K); the lower panel shows the location of the optimum DS on the potential barrier itself.

The correlation functions themselves (not shown) look

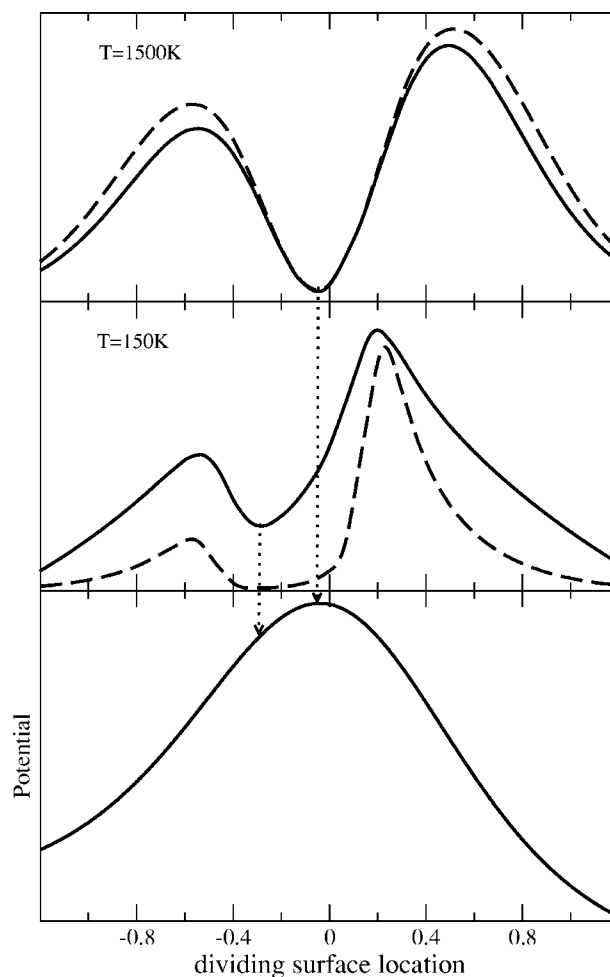


FIG. 3. Location of the dividing surface for the asymmetric Eckart potential barrier, determined by minimizing the highest-order coefficient in the log-augmented cumulant expansion. The solid and dashed lines show the value of this coefficient (the minimum of which determines the optimum dividing surface) for the two- and four-parameter models, respectively, for a high temperature ($T=1500$ K, upper panel) and a low temperature ($T=150$ K, middle panel). The lower panel shows the potential barrier and the location of the dividing surfaces on it.

very similar to those in Fig. 1. Figure 4 shows the percentage error, analogous to Fig. 2, for the two-, three-, and four-parameter versions of the model, as a function of temperature. One sees that all three versions are accurate to $\sim 5\%$ for $T > 200$ K, the error increasing to $\sim 15\%$ at 150 K.

Finally, it should be noted that the results for both the symmetric and asymmetric barriers, in Figs. 2 and 4, are in most cases a substantial improvement over those of the earlier QI model. The QI model (in its simplest one DS version) gives the rate to $\pm 25\%$ over this range of temperature. The one regime where the QI model is more accurate—at least in its most “rigorous” version that employs two DSs—is at very low temperature (“deep” tunneling).

IV. MULTIDIMENSIONAL EXAMPLES: THE COLLINEAR $\text{H} + \text{H}_2$ AND $\text{D} + \text{H}_2$ REACTIONS

A. $\text{H} + \text{H}_2$

In this case, like the 1d symmetric Eckart barrier, the optimum DS is determined by symmetry, the usual symmet-

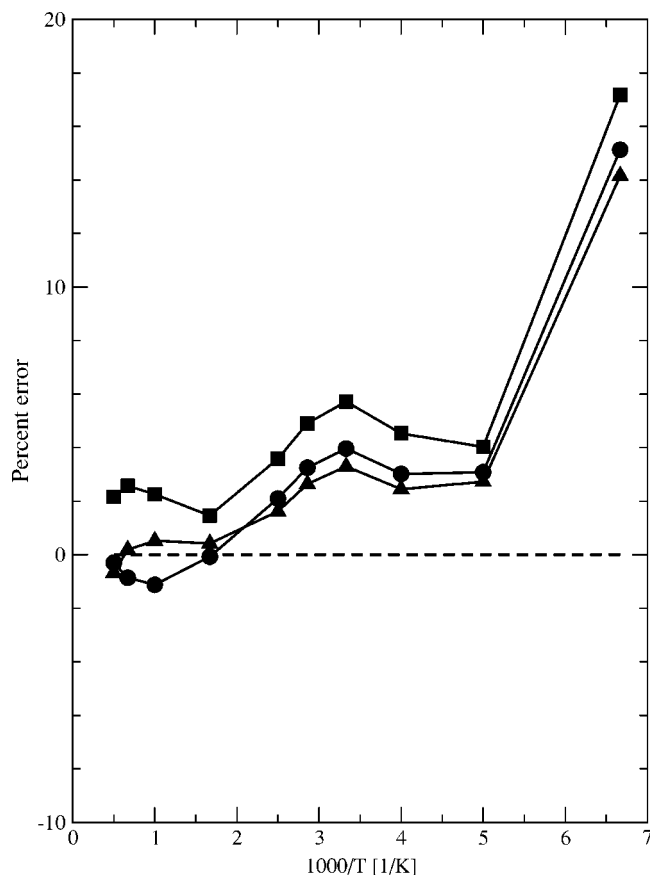


FIG. 4. Percentage error of the thermal rate constant vs temperature for the asymmetric Eckart barrier, where the various points refer to the same models as in Fig. 2.

ric dividing line in mass-weighted Cartesian coordinates. (We used the LSTH potential energy surface¹⁶ for these calculations.) In multidimensional problems, however, it is not possible in general to avoid recrossing flux no matter where the DS is located.^{17,18} This is well known to occur for the collinear $\text{H} + \text{H}_2$ reaction at high temperature,^{19,20} caused by high energy trajectories (speaking classically) that leave the DS and ricochet back across it.

Figure 5(a) shows the exact $C(t)$ at $T=2400$ K, compared to the two- and four-parameter models of Sec. II. Both of the latter are in excellent agreement with the correct $C(t)$ for short time, but they cannot describe the recrossing flux [negative region of $C(t)$] at longer time. At lower temperature, $T=400$ K, in Fig. 5(b) one sees that recrossing effects disappear, as is typically the case. We note also that recrossing effects are usually negligible for higher-dimensional systems.^{20–23} For example, even for this very elementary $\text{H} + \text{H}_2$ reaction, recrossing effects at very high temperature can lead to a factor of 2 error in the rate constant for the collinear version of the reaction, while they lead to only a 10% error in the rate for the $3d$ version of the reaction²⁰ (on the same potential energy surface).

Figure 6 shows the percentage error in the rate constant for the collinear $\text{H} + \text{H}_2$ reaction as a function of temperature, for the two-, three-, and four-parameter versions of the model of Sec. II. They are all accurate to $\sim 5\%$ until the high temperature regime where recrossing effects cause these

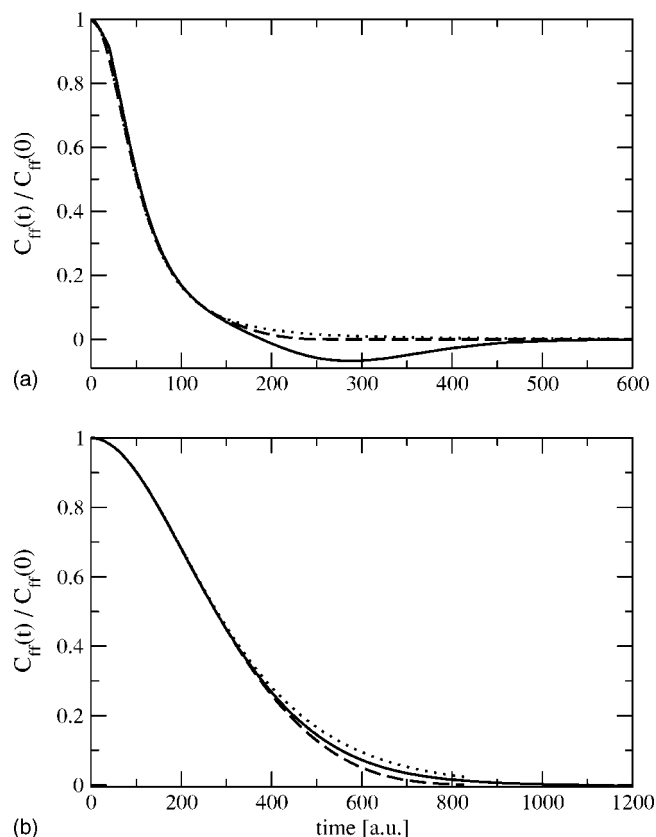


FIG. 5. Flux-flux autocorrelation function vs time for the collinear $\text{H} + \text{H}_2$ reaction at (a) $T=2400$ K and (b) $T=400$ K using the same models as in Fig. 1.

transition statelike models to be too large. As noted in the previous paragraph, this effect is usually of minimal relevance for reactions in complex systems.

B. $\text{D} + \text{H}_2$

Finally, we consider the asymmetric version of this reaction, collinear $\text{D} + \text{H}_2 \rightarrow \text{DH} + \text{H}$. We used the TK potential energy surface¹⁶ and performed the calculations in mass-scaled normal mode coordinates where, because of the deuterium mass, the potential energy surface is asymmetric and consequently the location of the DS is no longer determined by symmetry.

As for the $1d$ asymmetric barrier, the location of the dividing surfaces was chosen by minimizing the highest-order parameter in the cumulant expansion. Figure 7 shows the optimum linear²⁴ DS thus determined for three different temperatures. At the highest temperature (1500 K, dotted line) it passes through the saddle point of the potential energy surface and is essentially the standard classical result (the symmetric stretch as the saddle point). As the temperature is lowered (solid line for 300 K, dashed line for 200 K), the DS moves farther from the saddle point and rotates to be approximately perpendicular to the minimum energy path. This behavior is clearly understood from conventional theory (based on the reaction path Hamiltonian^{25,26}), but here it emerges “automatically” from the criterion described above. Figure 8 shows the correlation functions obtained with these optimum dividing surfaces for the two- and four-parameter

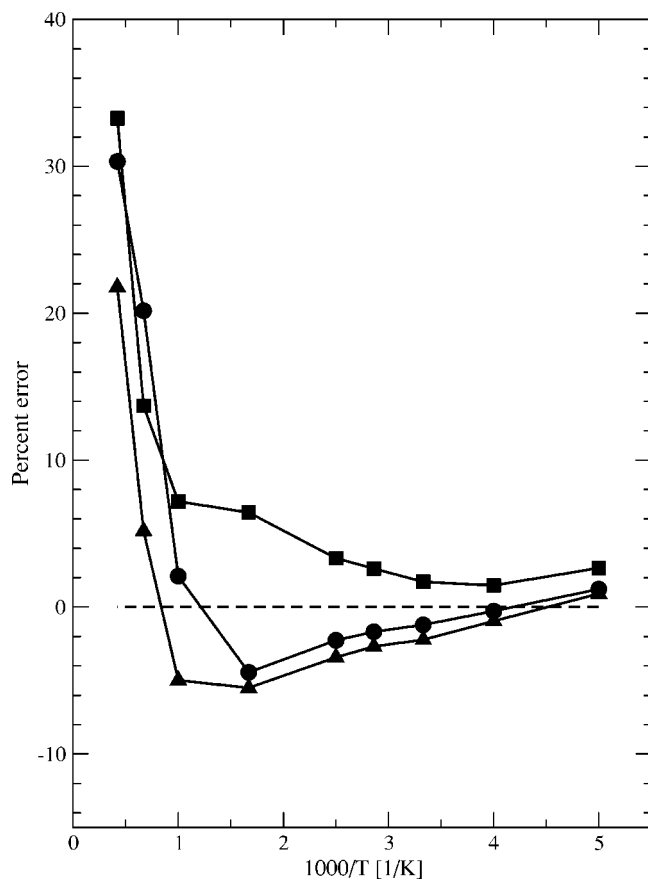


FIG. 6. Percentage error of the thermal rate constant for the collinear reaction $\text{H} + \text{H}_2$ vs temperature using the same models as in Fig. 2. Exact QM values from Ref. 27.

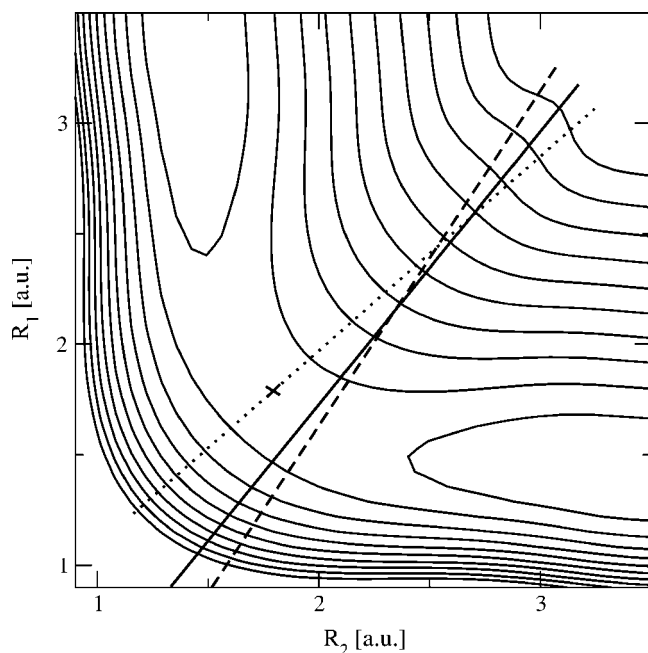


FIG. 7. Optimum dividing surface for the collinear $\text{D} + \text{H}_2$ reaction (shown on the contour plot of the potential energy surface); X marks the saddle point. The dotted line is for $T = 1500$ K, the solid line for $T = 300$ K, and the dashed line for $T = 200$ K.

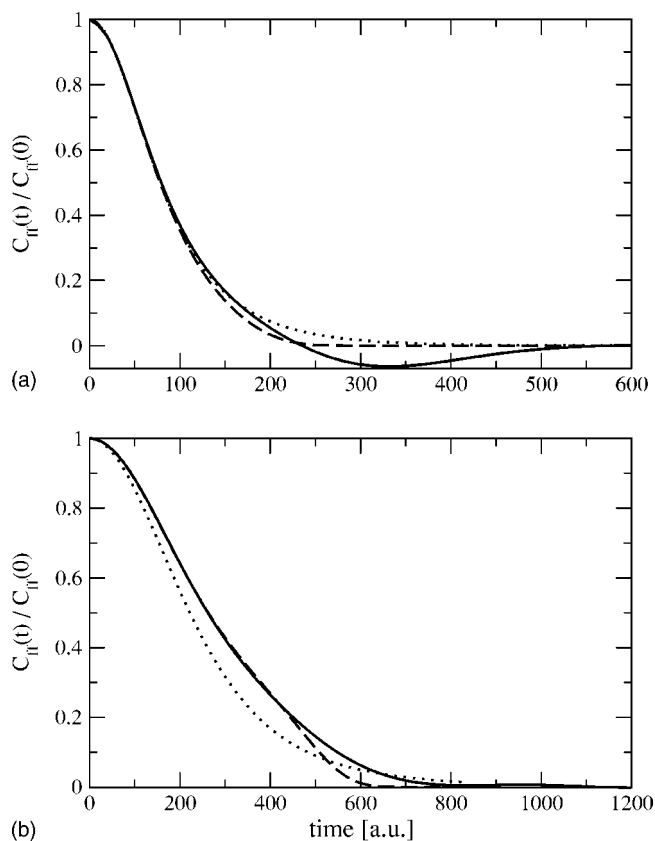


FIG. 8. Flux-flux autocorrelation function vs time for the collinear reaction $\text{D} + \text{H}_2$ at (a) $T = 1500$ K and (b) $T = 400$ K using the same models as in Fig. 1.

models for a high temperature [1500 K, Fig. 8(a)] and a low temperature [400 K, Fig. 8(b)]; again, one sees recrossing effects at high temperature which are typically unimportant for systems with more degrees of freedom.

Figure 9 shows the percentage error in the rate constant for the $\text{D} + \text{H}_2$ reaction as a function of temperature for the two-, three-, and four-parameter versions of the present model. As before, the error is within $\sim 5\%$ except for the high temperature region where recrossing effects become significant in this collinear example.

V. SUMMARY AND CONCLUSIONS

The purpose of this paper has been to present an extension of the quantum instanton model¹ for chemical reaction rates, one that is still expressed solely in terms of the (quantum) Boltzmann operator but which is more accurate and thus potentially more useful than the original QI theory. It uses the low-order zero time derivatives of the MST flux-flux autocorrelation function⁹—which can be efficiently computed by Monte Carlo methods² for large molecular systems by the methods of Predescu and Doll⁸—to express the short time correlation function by a log-augmented cumulant expansion. The rate is then given by the integral of this approximate correlation function.

This model can be viewed as a type of “quantum transition state theory” since it provides a short time approximation for the flux correlation that does not describe any effects of recrossing flux. For activated reactions in systems with

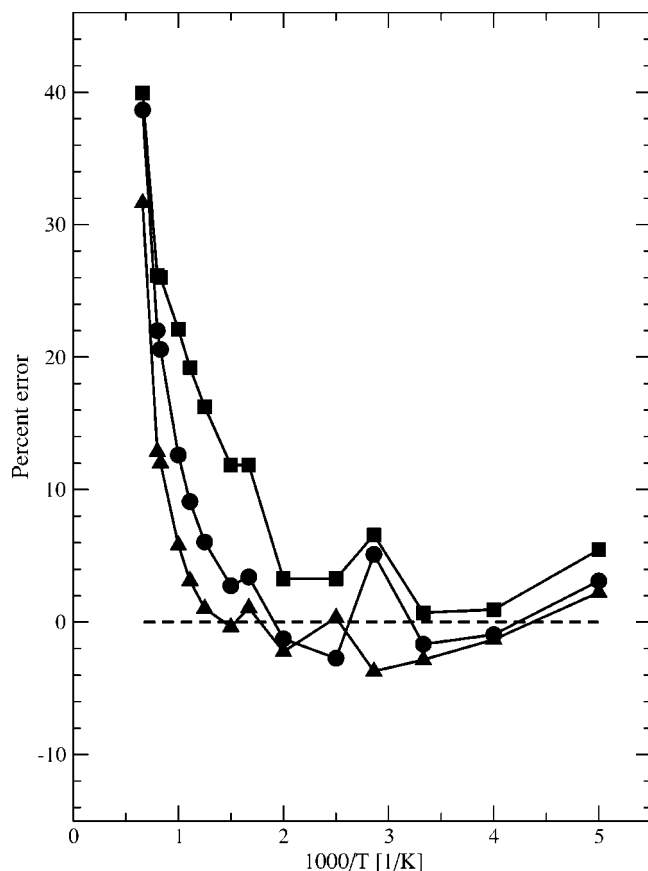


FIG. 9. Percentage error of the thermal rate constant for the collinear $D + H_2$ reaction vs temperature using the same models as in Fig. 2. Exact QM values from Ref. 28.

many degrees of freedom, this is usually not a significant limitation, but it does require that one choose an appropriate dividing surface (which defines the flux operator). A general prescription was suggested for choosing the optimum DS which was seen to provide excellent results for the examples treated here. (It works much better, for example, than choosing the DS to minimize the rate constant itself, which is the prescription of classical transition state theory.) Further work on providing a more rigorous and well-defined way of choosing the optimum dividing surface, however, would be useful.

For the $1d$ and collinear reactions treated here, both symmetric and asymmetric, this model is a considerable improvement over the original quantum instanton theory, giving the rate for these examples to within $\sim 5\%$ – 10% over a wide range of temperature (except for high temperature for the collinear reactions where recrossing effects are significant). It is essentially as easy to implement as the original QI model and thus should be useful in a variety of applications.

ACKNOWLEDGMENTS

This work was supported by the Director, Office of Science, Office of Basic Energy Sciences, Chemical Sciences, Geosciences, and Biosciences Division, U.S. Department of Energy under Contract No. DE-AC03-76SF00098 and by the National Science Foundation Grant No. CHE-0345280.

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